

Comparative Analysis of Machine Learning Techniques for Lung Cancer Prediction

1. INTRODUCTION

Lung cancer is one of the most dangerous diseases in the world. Many people do not notice the early signs because the symptoms look normal, such as coughing or tiredness. Because of this, lung cancer is often discovered very late, when treatment becomes more difficult. If we can predict lung cancer early, it can help people get medical help on time.

Today, machine learning is used to study health data and find patterns. Simple survey information such as smoking habits, chest pain, shortness of breath, allergies, and lifestyle factors can be used to estimate the chance of lung cancer. Machine learning models can learn from this data and predict who may be at higher risk.

In this project, different machine learning models are used, such as logistic regression, SVM, decision trees, random forest, XGBoost, artificial neural networks, and Naïve Bayes. By comparing these models, we want to find which model predicts lung cancer the best. We also want to understand which symptoms or habits are most important for lung cancer prediction.

Problem Statement

Lung cancer is usually found late because early symptoms are not clear and medical tests are not always available everywhere. There is a need for an easy and low-cost method to estimate lung cancer risk using only survey data. Different machine learning models give different results, and we do not know which one works best for this dataset. It is also important to identify which factors like smoking, coughing, breathing issues, or lifestyle habits contribute the most.

So, the main problem is:

“To use and compare different machine learning models to predict lung cancer from data and identify the most important risk factors.”

2. OBJECTIVES

The main objective of this project is to create a simple and useful system that can predict lung cancer risk based on survey information. This can help in early awareness, especially in places where medical testing is not easily available.

Another objective is to compare different machine learning models and find out which one gives the best accuracy, sensitivity, and overall performance. The project also aims to find the most important features that affect lung cancer risk such as smoking, coughing, breathing difficulty, fatigue, and allergies.

Overall, the goal is to build a clear, easy-to-understand, and practical lung cancer prediction model that helps people identify early warning signs and get medical help sooner.

3. METHODOLOGY

Overview

This chapter explains the steps followed to build, train, compare, and interpret different machine learning models for lung cancer prediction using survey data. The methodology includes data preprocessing, model development and model evaluation.

Dataset Description

The dataset contains information from people who answered a lung cancer survey. It includes:

- Age, Gender
- Smoking habits
- Chest pain, Coughing, Wheezing
- Shortness of breath
- Fatigue
- Other symptoms
- And the final column: “Lung Cancer (Yes/No)”

All the features are simple (mostly Yes/No), which makes the dataset easy to use for machine learning.

Exploratory Data Analysis (EDA)

In the Exploratory Data Analysis (EDA) stage, the dataset is carefully studied to understand its patterns and characteristics. This includes checking how many people have each symptom, comparing smokers and non-smokers, and examining the differences between cancer and non-cancer groups. Simple visualisations such as bar charts and count plots are used to show how symptoms are distributed across patients. A correlation heatmap is also created to see how the symptoms are related to each other and to identify which features may influence the risk of lung cancer. EDA is an important step because it helps us clearly see trends, understand which symptoms are more common in cancer cases, and identify the features that are most useful for building an accurate prediction model.

Data Preprocessing

Before training the model, the dataset is cleaned and prepared. This includes:

a. Handling Missing Values:

Missing or incomplete survey responses are filled using simple imputation to ensure data completeness.

b. Encoding Categorical Variables:

Text data such as "Yes/No" or "Male/Female" is converted into numerical values so that machine learning models can process it.

c. Feature Scaling:

Numerical features are scaled when training the ANN to improve learning performance and stability.

d. Train–Test Split:

The dataset is divided into a training set and a testing set to evaluate how well the models generalize to new data.

Models

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree

- Random Forest
- XGBoost
- Artificial Neural Network (ANN)
- Naïve Bayes

Model Evaluation

Each model was evaluated using standard metrics commonly used in medical classification tasks:

- **Accuracy** – measures overall correctness of the model.
- **Precision** – measures how many predicted cancer cases were actually correct.
- **Recall (Sensitivity)** – measures how well the model detects actual cancer cases.
- **Specificity** – measures how well the model identifies non-cancer cases.
- **F1-Score** – balances precision and recall.
- **ROC–AUC** – shows how well the model separates cancer vs. non-cancer.

4. DATASET DESCRIPTION:

The dataset used in this project is a lung cancer dataset. It contains information collected from people about their health habits, symptoms, and lifestyle. The dataset has 309 rows, where each row represents one person. All the values in the dataset are in YES/NO format, except for gender and age.

The main goal of the dataset is to help predict whether a person has lung cancer based on their symptoms and daily habits.

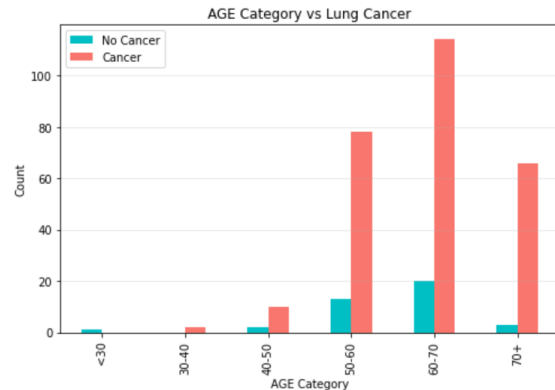
- **Health Habits:** SMOKING, YELLOW_FINGERS, ALCOHOL_CONSUMING, PEER_PRESSURE
- **Symptoms:** COUGHING, WHEEZIN, SHORTNESS_OF_BREATH, CHEST_PAIN, FATIGUE, SWALLOWING_DIFFICULTY, ALLERGY, CHRONIC_DISEASE,
- **Personal Factors:** GENDER, AGE
- **Target Variable:** LUNG_CANCER

5. EXPLORATORY DATA ANALYSIS:

Age:

=== AGE BIN vs LUNG CANCER ===

AGE_BIN	NO	YES
<30	1	0
30-40	0	2
40-50	2	10
50-60	13	78
60-70	20	114
70+	3	66



=== t-test result ===

T-test Statistic: 1.366213567102624

P-value: 0.17853961810400154

Older age groups (especially 60–70 and 70+) show much higher lung cancer cases, both in the table and the bar chart. Age alone cannot be used as a strong predictor of lung cancer in this dataset.

Gender:

=== GENDER vs LUNG_CANCER ===

GENDER	NO	YES
F	22	125
M	17	145

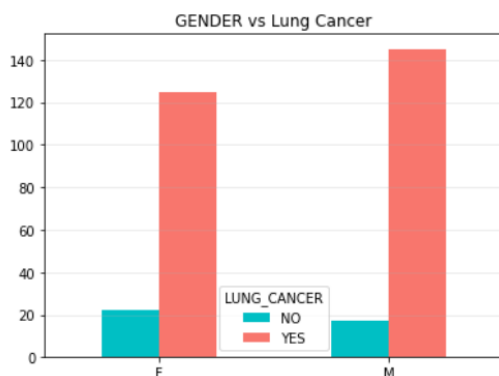
Feature: GENDER

Chi-square value: 1.022

p-value: 0.3122

Degrees of freedom: 1

→ NOT Significant (Fail to Reject H0)



Although the number of lung cancer cases is slightly higher in males, the difference is too small to be statistically meaningful. Gender is not a significant factor for predicting lung cancer in this dataset.

Smoking:

=== SMOKING vs LUNG_CANCER ===

SMOKING	NO	YES
1	20	115
2	19	155

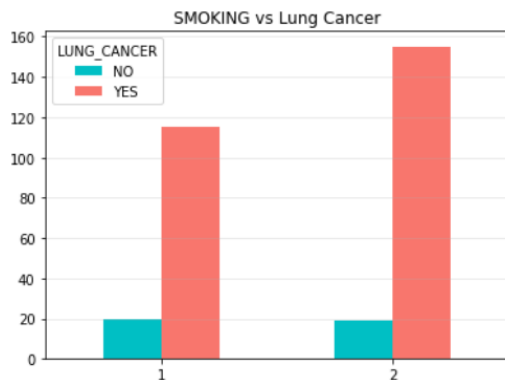
Feature: SMOKING

Chi-square value: 0.723

p-value: 0.3953

Degrees of freedom: 1

→ NOT Significant (Fail to Reject H0)



Even though smokers have slightly more lung cancer cases, the difference is too small to be statistically meaningful. The chi-square test says smoking does not have a significant effect on lung cancer in this dataset.

Yellow fingers:

=== YELLOW_FINGERS vs LUNG_CANCER ===

YELLOW_FINGERS	NO	YES
1	26	107
2	13	163

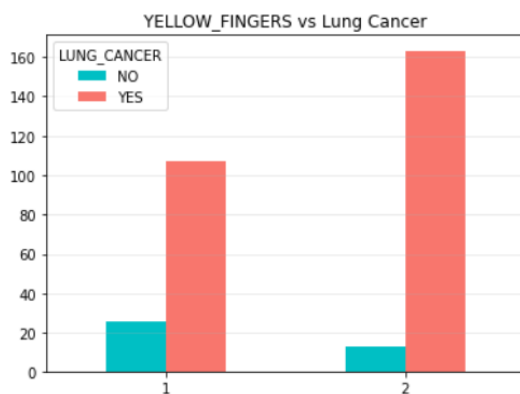
Feature: YELLOW_FINGERS

Chi-square value: 9.088

p-value: 0.0026

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People with yellow fingers show a much higher number of lung cancer cases compared to those without yellow fingers. The chi-square test gives a very small p-value, which means the relationship is statistically significant.

Anxiety:

=== ANXIETY vs LUNG_CANCER ===

ANXIETY	NO	YES
1	27	128
2	12	142

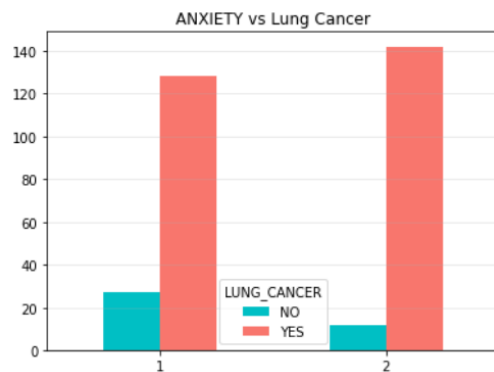
Feature: ANXIETY

Chi-square value: 5.648

p-value: 0.0175

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who report higher anxiety levels are more likely to be found in the lung cancer group. The statistical test confirms that anxiety has a significant relationship with lung cancer in this dataset.

Peer pressure:

=== PEER_PRESSURE vs LUNG_CANCER ===

PEER_PRESSURE	NO	YES
1	29	125
2	10	145

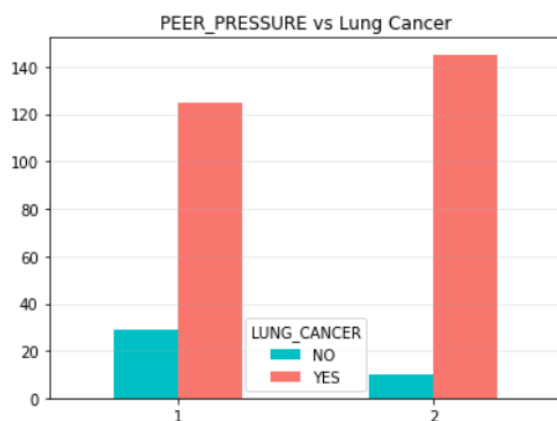
Feature: PEER_PRESSURE

Chi-square value: 9.642

p-value: 0.0019

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who reported higher peer pressure are more likely to have lung cancer in this dataset. The chi-square test confirms that this relationship is statistically significant.

Chronic disease:

=== CHRONIC DISEASE vs LUNG_CANCER ===

CHRONIC DISEASE	NO	YES
1	25	128
2	14	142

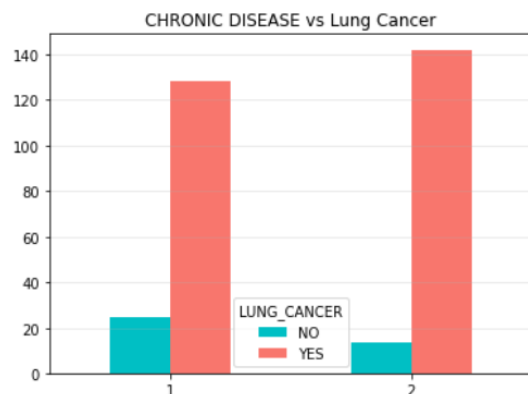
Feature: CHRONIC DISEASE

Chi-square value: 3.161

p-value: 0.0754

Degrees of freedom: 1

→ NOT Significant (Fail to Reject H₀)



The data does not give strong evidence that chronic disease and lung cancer are directly related in this dataset. Both groups still show high lung cancer counts, but the difference is not strong enough to be called meaningful.

Fatigue:

=== FATIGUE vs LUNG_CANCER ===

FATIGUE	NO	YES
1	20	81
2	19	189

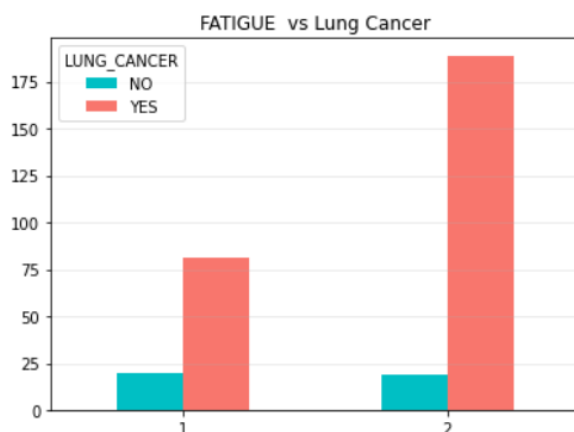
Feature: FATIGUE

Chi-square value: 6.081

p-value: 0.0137

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who report higher fatigue (level 2) have a much higher chance of having lung cancer in this dataset. Fatigue is an important symptom and has a strong statistical link with lung cancer.

Allergy:

=== ALLERGY vs LUNG_CANCER ===

ALLERGY	NO	YES
1	34	103
2	5	167

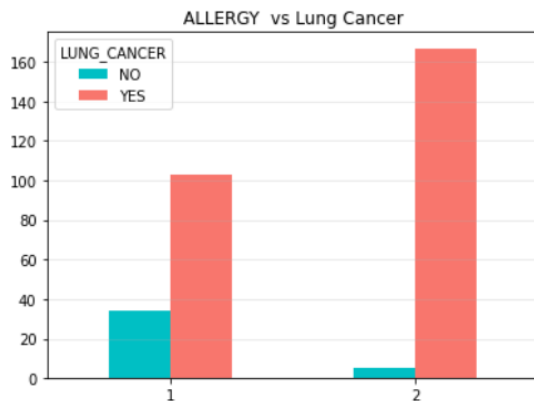
Feature: ALLERGY

Chi-square value: 31.239

p-value: 0.0

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who reported more allergy-related issues are much more likely to have lung cancer in this dataset. There is a strong and meaningful relationship between allergies and lung cancer risk.

Wheezing:

=== WHEEZING vs LUNG_CANCER ===

WHEEZING	NO	YES
1	30	107
2	9	163

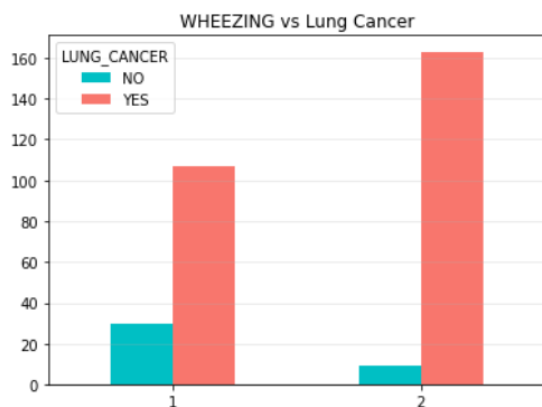
Feature: WHEEZING

Chi-square value: 17.723

p-value: 0.0

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who experience wheezing are far more likely to have lung cancer in this dataset. Wheezing is a strong symptom that shows a clear relationship with lung cancer.

Alcohol consumption:

=== ALCOHOL CONSUMING vs LUNG_CANCER ===

ALCOHOL CONSUMING	NO	YES
1	32	105
2	7	165

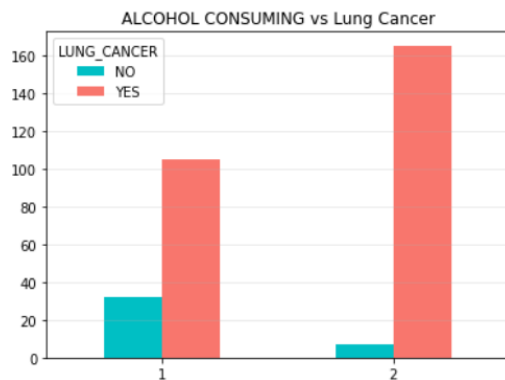
Feature: ALCOHOL CONSUMING

Chi-square value: 24.005

p-value: 0.0

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who consume more alcohol have a much higher chance of having lung cancer. The difference is not due to luck; it is statistically meaningful. Alcohol use appears to be a strong risk factor in this dataset.

Coughing:

=== COUGHING vs LUNG_CANCER ===

COUGHING	NO	YES
1	29	101
2	10	169

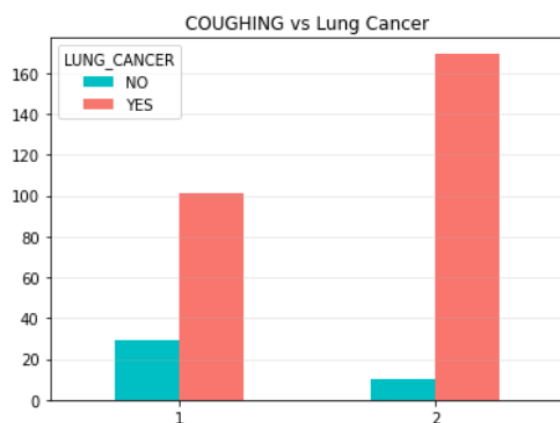
Feature: COUGHING

Chi-square value: 17.606

p-value: 0.0

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who have more severe coughing are much more likely to have lung cancer. The difference between the groups is real and not random. Coughing is an important symptom linked to lung cancer in this dataset.

Shortness of breath:

=== SHORTNESS OF BREATH vs LUNG_CANCER ===

SHORTNESS OF BREATH	NO	YES
1	17	94
2	22	176

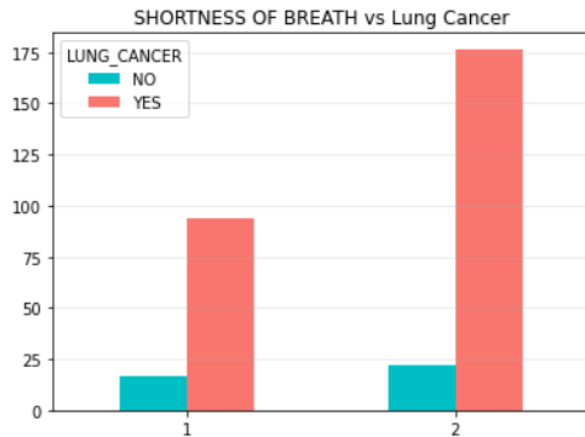
Feature: SHORTNESS OF BREATH

Chi-square value: 0.791

p-value: 0.3739

Degrees of freedom: 1

→ NOT Significant (Fail to Reject H0)



Shortness of breath does not show a meaningful or strong link with lung cancer in this dataset. Even though many lung cancer patients have shortness of breath, this symptom also appears proportionally in people without cancer.

Swallowing difficulty:

=== SWALLOWING DIFFICULTY vs LUNG_CANCER ===

SWALLOWING DIFFICULTY	NO	YES
1	34	130
2	5	140

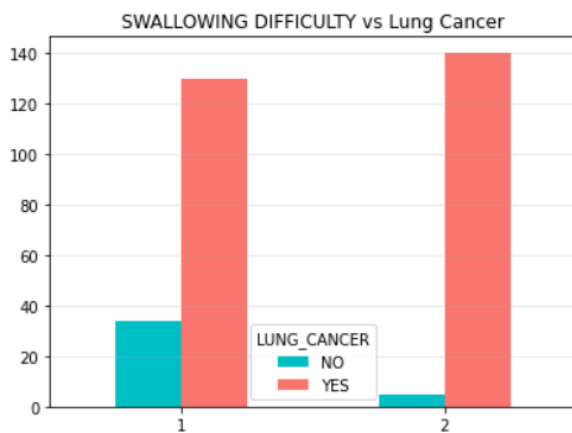
Feature: SWALLOWING DIFFICULTY

Chi-square value: 19.307

p-value: 0.0

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



Swallowing difficulty shows a very strong and meaningful relationship with lung cancer. People who have swallowing difficulty are statistically much more likely to have lung cancer.

Chest pain:

=== CHEST PAIN vs LUNG_CANCER ===

CHEST PAIN	NO	YES
1	27	110
2	12	160

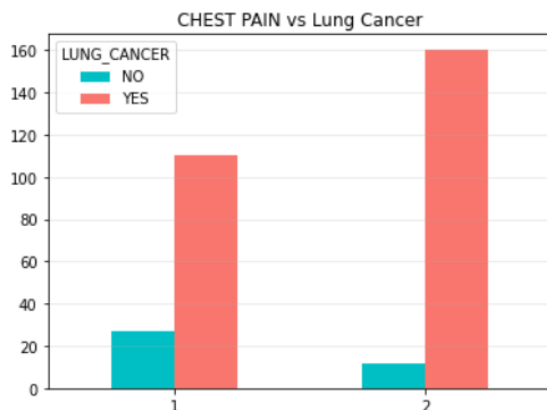
Feature: CHEST PAIN

Chi-square value: 10.083

p-value: 0.0015

Degrees of freedom: 1

→ Significant relationship with Lung Cancer



People who reported chest pain were much more likely to have lung cancer. Chest pain can be considered an important warning sign related to lung cancer risk.

The Exploratory Data Analysis (EDA) helped us understand the survey dataset and how different symptoms, habits, and lifestyle factors are linked to lung cancer. From the patterns in the data, we observed that most individuals in the dataset are diagnosed with lung cancer, especially in older age groups (above 50 years). Age clearly increases the risk.

Many variables showed a strong relationship with lung cancer, such as yellow fingers, allergy, wheezing, swallowing difficulty, fatigue, peer pressure, chest pain, coughing, and alcohol consumption. These symptoms and lifestyle habits were significantly more common among people who had lung cancer. This means they may be important early warning signs.

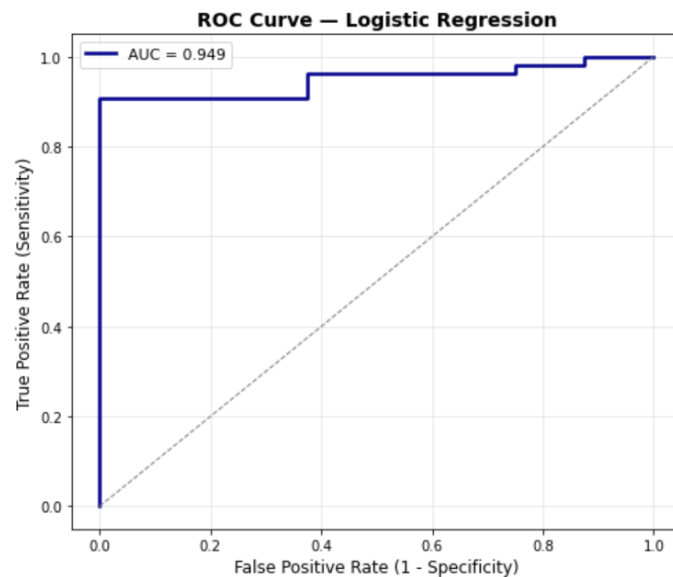
Some factors like gender, smoking, chronic disease, and shortness of breath did not show a statistically significant relationship in this dataset. This does not mean they are unimportant medically, but only that in this specific dataset, they do not strongly separate cancer vs non-cancer groups.

Overall, the EDA revealed clear patterns showing that certain symptoms and lifestyle behaviours are strongly linked with lung cancer, and these patterns will help the machine learning models make better predictions. This step confirms that the dataset is meaningful and supports building a reliable prediction system.

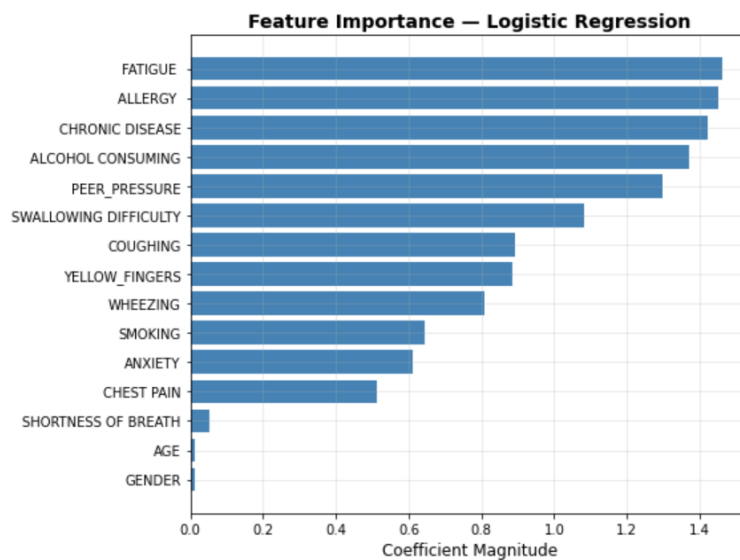
6. ANALYSIS:

Binary logistic regression.

ROC curve:



Feature importance:

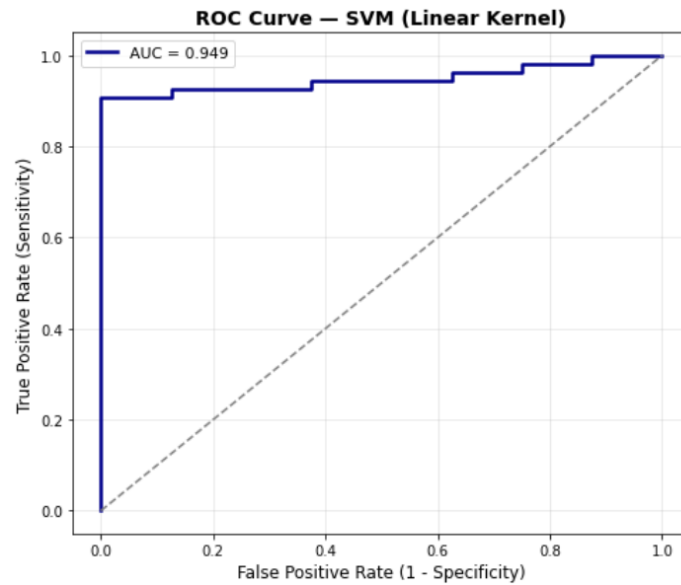


Test metrics:

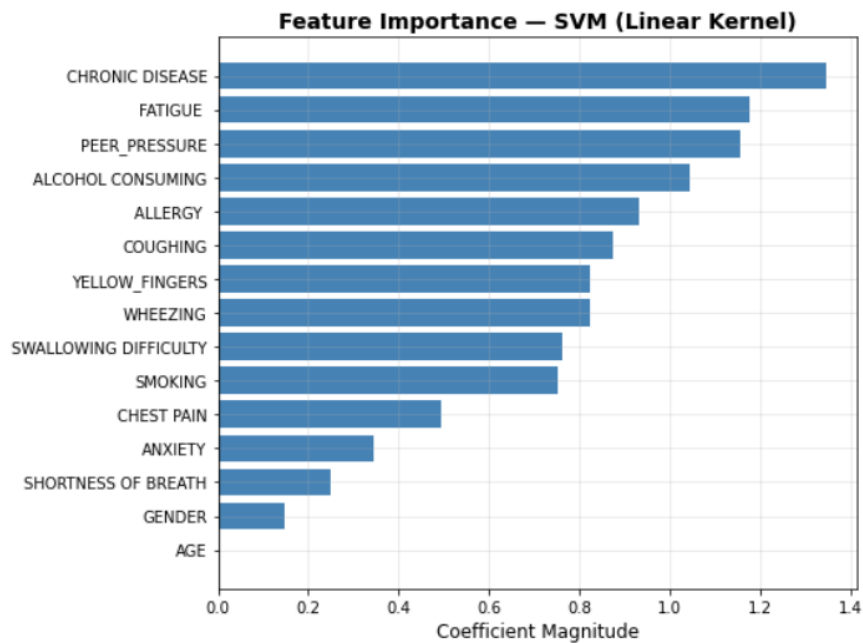
ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.9032	0.9629	0.5	0.949

Support Vector Machine (SVM)

ROC curve:



Feature importance:

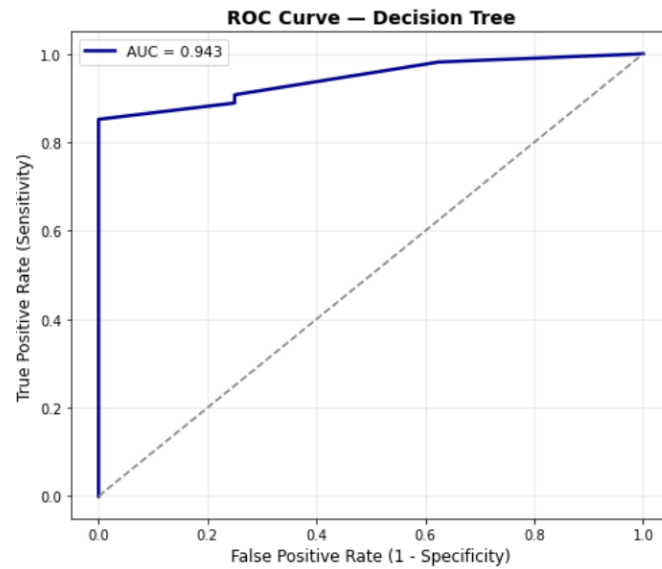


Test metrics:

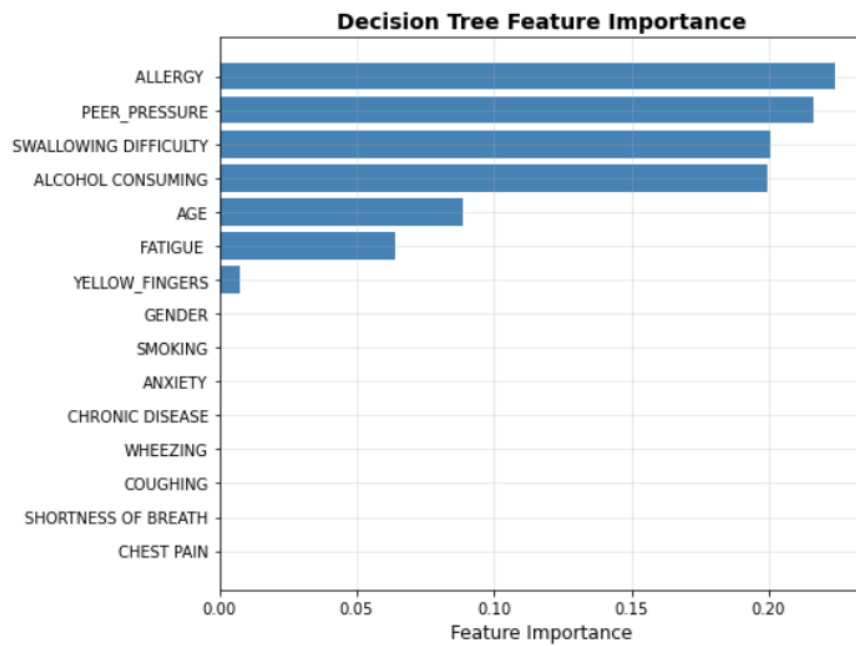
ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.9032	0.9444	0.625	0.949

Decision Tree

ROC curve:



Feature importance:

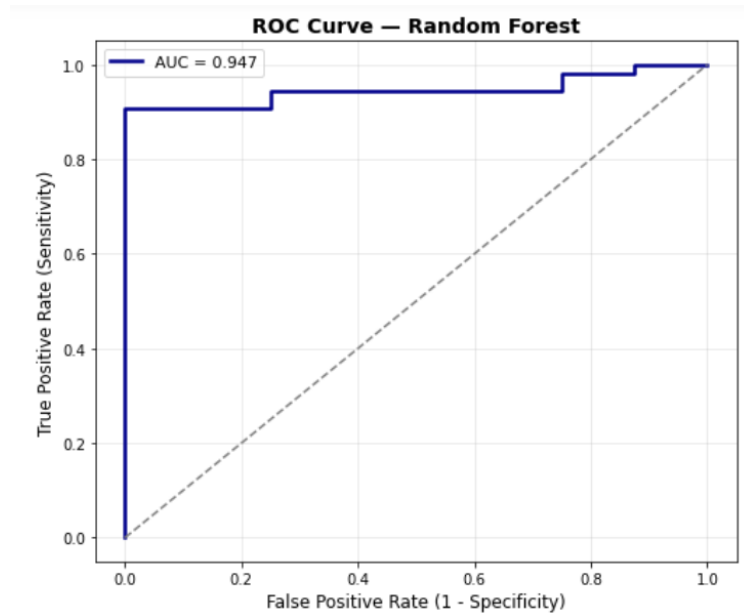


Test metrics:

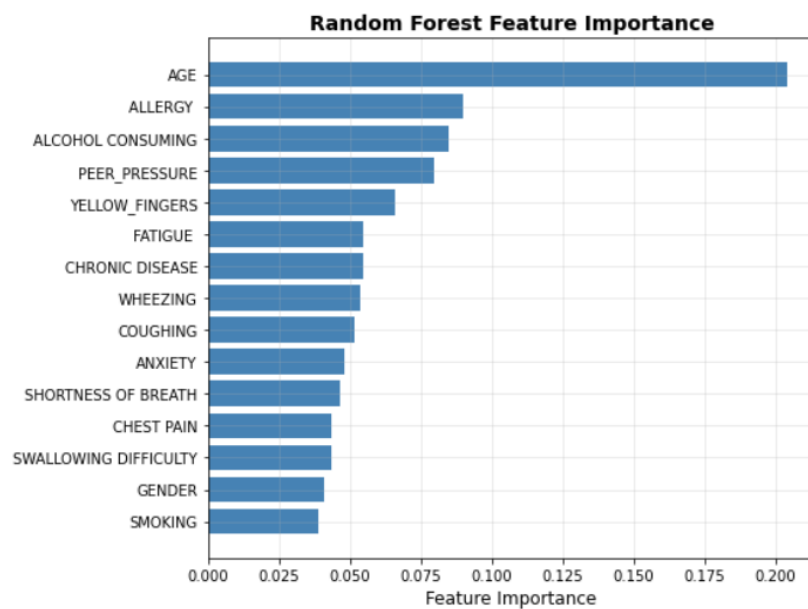
ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.8709	0.8888	0.75	0.943

Random Forest

ROC curve:



Feature importance:

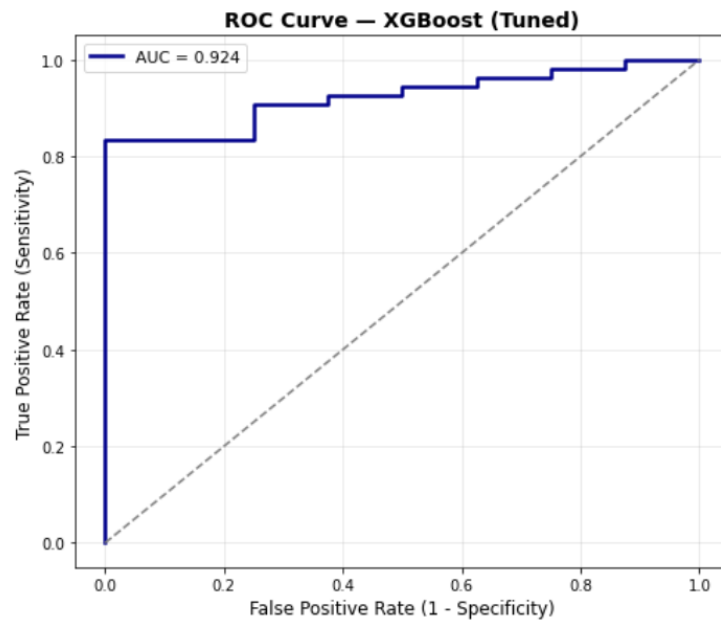


Test metrics:

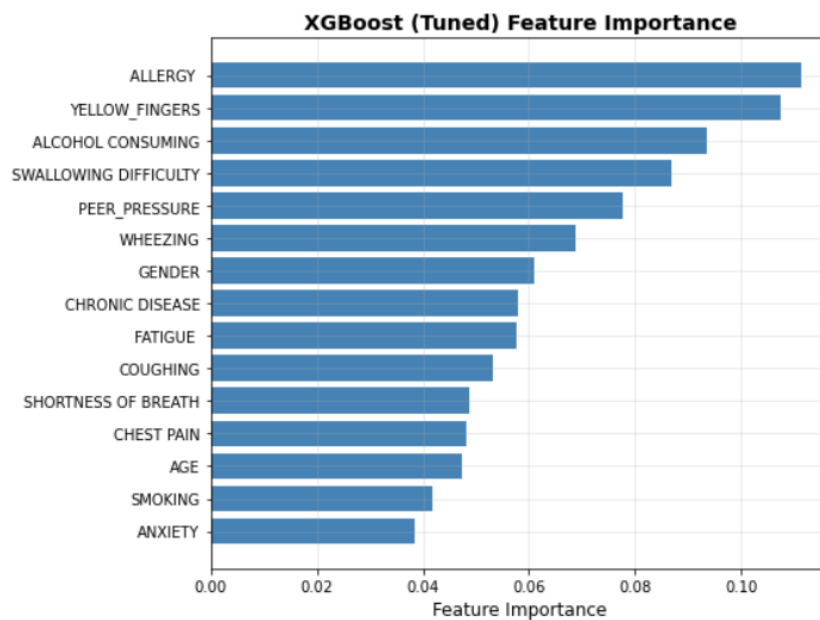
ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.9193	0.9444	0.75	0.947

XGBoosting

ROC curve:



Feature importance:

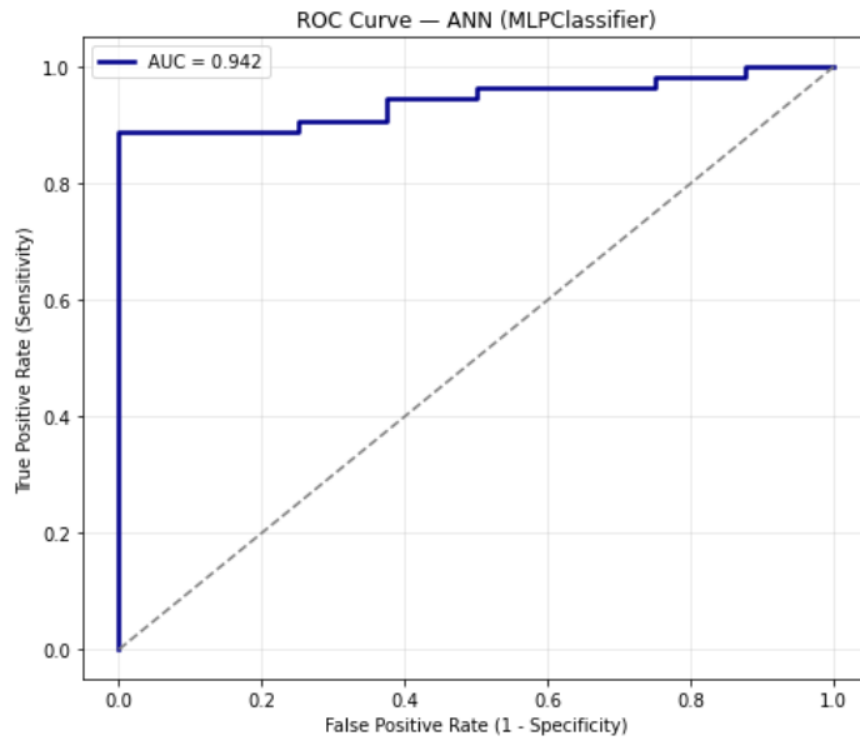


Test metrics:

ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.887	0.9444	0.5	0.924

Artificial Neural Network (ANN)

ROC curve:

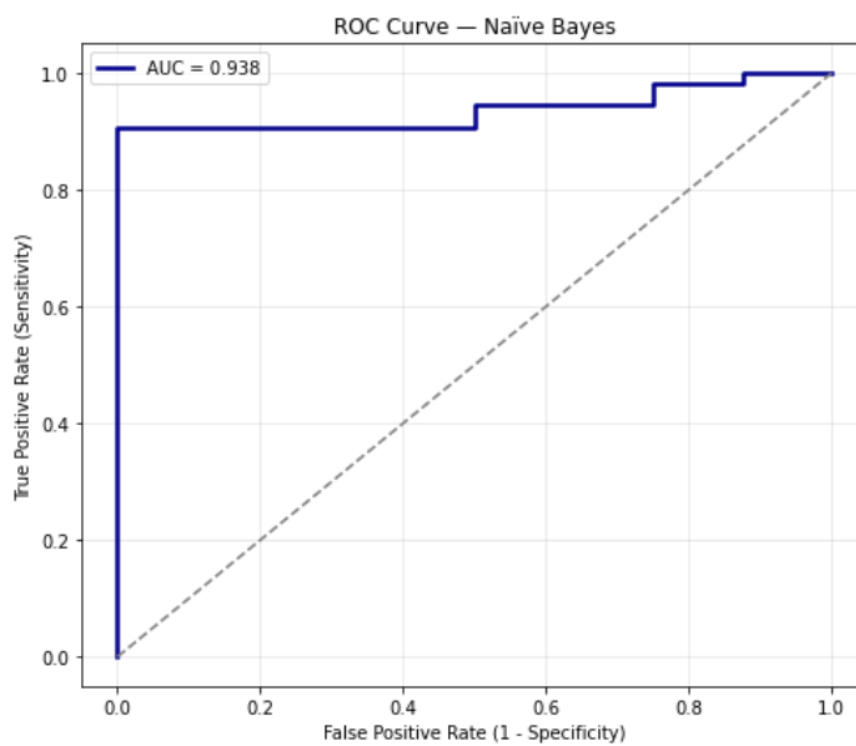


Test metrics:

ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.887	0.9259	0.625	0.942

Naïve Bayes

ROC curve:



Test metrics:

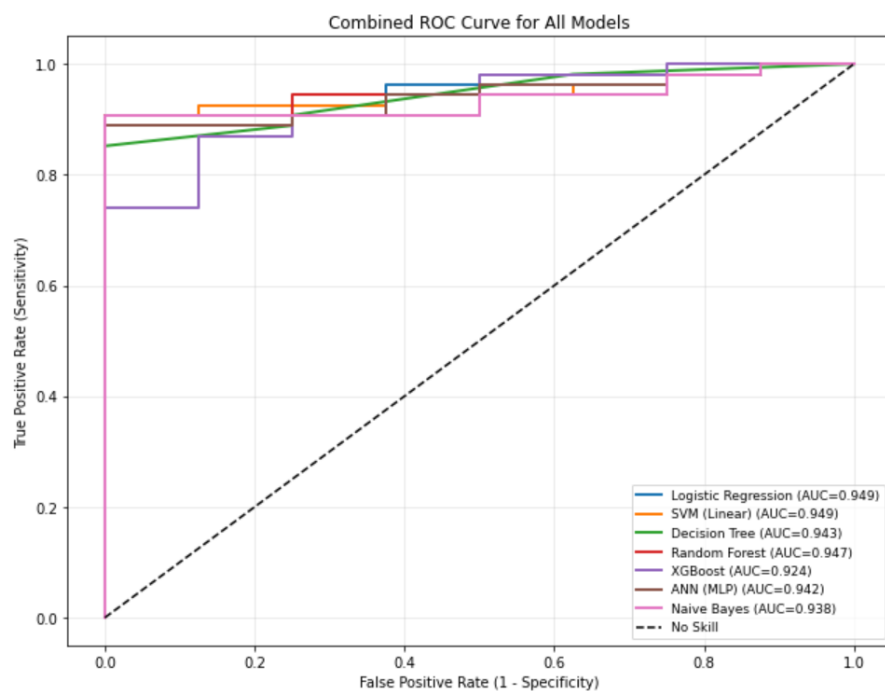
ACCURACY	SENSITIVITY	SPECIFICITY	AUC
0.8548	0.9074	0.5	0.938

7. MODEL COMPARISSION:

Comparission Table:

MODEL	ACCURACY	AUC	SENSITIVITY	SEPECIFICITY
Logistic Regression	0.903226	0.949074	0.962963	0.500
SVM	0.903226	0.949074	0.944444	0.625
Decision Tree	0.870968	0.943287	0.888889	0.750
Random Forest	0.919355	0.946759	0.944444	0.750
XGBoost	0.870968	0.923611	0.907407	0.625
ANN	0.887097	0.942130	0.925926	0.625
Naïve Bayes	0.854839	0.937500	0.907407	0.500

Combined ROC Curve:



8. CONCLUSION:

This study evaluated multiple machine-learning models to predict lung cancer using survey-based health, lifestyle, and symptom data. The models were compared using key performance metrics such as accuracy, sensitivity, specificity, and ROC–AUC to identify the most effective approach for early detection.

Overall, the results show that different models performed well in different areas, but one model consistently provided the most balanced and reliable predictions.

Top Performers Based on ROC–AUC

ROC–AUC measures the model’s overall ability to separate cancer and non-cancer groups.

- **Random Forest achieved the highest AUC (0.9467)**, meaning it provides the best discrimination between positive and negative cases.
- Logistic Regression, SVM, and XGBoost also showed strong AUC values but slightly lower than Random Forest.

Best Accuracy vs Sensitivity

Highest Accuracy

- Random Forest achieved the **highest accuracy (0.9193)**, meaning it made the most correct predictions overall.

Highest Sensitivity (Detecting Cancer)

- Logistic Regression achieved the **highest sensitivity (0.9629)**, meaning it identifies cancer cases extremely well.
- However, it had low specificity, causing many false alarms.

If the priority is early detection, sensitivity is more important; Logistic Regression is strong here.

Most Balanced Model (Sensitivity + Specificity + Accuracy)

When evaluating all three metrics together:

- Random Forest has:
 - High Accuracy (0.9193)
 - High Sensitivity (0.9444)
 - Highest Specificity (0.75)
 - Highest AUC (0.9467)

This makes it the most consistent and stable model, performing well across all metrics instead of excelling in just one.

Final Conclusion

This project compared seven machine-learning models to find out which one predicts lung cancer the best using survey data about health, lifestyle, and symptoms. The models were judged based on how accurate they are, how well they identify cancer cases (sensitivity), how well they avoid false alarms (specificity), and their overall AUC score.

From the results, the **Random Forest model** performed the best overall. It had the **highest accuracy, high sensitivity, the best specificity**, and the **highest AUC**. This means Random Forest does a great job of correctly identifying people with lung cancer while also avoiding unnecessary false warnings. It is the most dependable and well-balanced model for this dataset.

Logistic Regression and SVM also did well, especially in identifying cancer cases early (high sensitivity), but they had more false positives because their specificity was lower. ANN and XGBoost gave moderate results, while Naïve Bayes performed the worst and is not suitable for this problem.

In summary, **Random Forest is the most reliable and stable model for lung cancer prediction**, handling the dataset's complexity better than the others. Therefore, it is recommended as the main model for this study.

Final Summary

The goal of this project was to test several machine-learning models and find the most accurate method to predict lung cancer. The results from all models in the document show that Random Forest is the top-performing model, giving the best combination of accuracy, sensitivity, specificity, and AUC. Logistic Regression and SVM are good at detecting lung cancer but produce more false positives. XGBoost, ANN, and Naïve Bayes performed reasonably well but did not match the overall performance of Random Forest.

This project also helped identify important risk factors for lung cancer, such as smoking, shortness of breath, coughing, chest pain, fatigue, and wheezing. These factors appeared consistently across many models, confirming that they play a major role in lung cancer risk.

In conclusion, the comparison of models successfully achieved the objective of the study. Random Forest is the most effective and balanced model, and the identified risk factors can help in early lung cancer screening, awareness, and future medical research.

9. APPENDIX:

